Lab 7: Fuzzing for Input Validation Bugs

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Assigned repo: L6_L7_14_library_fines (Lab 7) Tools: Python 3.11.9, pytest 9.2.0, Hypothesis 6.170.15

Step 2: Run the buggy application

python library_fines.py with a valid input file, then an invalid one. The invalid member ID crashes the program with a ValueError.

```
PS D:\SE-Lab\Lab7> Get-Content inputs\valid.txt; cmd /c "python library_fines.py < inputs\valid.txt"; Get-Content
inputs\invalid.txt; cmd /c "python library_fines.py < inputs\invalid.txt"
STU-1023
10,20,30
Data Structures
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): Member Type and Number: ('STU', 1023)

==== Running total_fine ====
Enter days kept for each book as CSV (e.g. 10,20,30): Total Fine: 44

==== Running shelf_code ====
Enter a book title with at least two words: Shelf Code: DS
STU1023
10,,30
Python
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): Traceback (most recent call last):
  File "D:\SE-Lab\Lab7\library_fines.py", line 34, in <module>
    main()
  File "D:\SE-Lab\Lab7\library_fines.py", line 20, in main
    result = parse_member_id(user_input)
             ^^^^^^^^^^^^^^^^^^^^^^^^^^^
  File "D:\SE-Lab\Lab7\library_fines.py", line 2, in parse_member_id
    member_type, number = member_id.split('-')
    ^^^^^^^^^^^^^^^^^^
ValueError: not enough values to unpack (expected 2, got 1)
```

Step 4: Run the Hypothesis tests on the buggy code

pytest test_library_fines.py - 4 failed, 3 passed. Hypothesis found 7 distinct crashes and that negative days are accepted.

```
PS D:\SE-Lab\Lab7> pytest test_library_fines.py
============================================= test session starts =============================================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\SE-Lab\Lab7
plugins: hypothesis-6.170.15, platformdirs-4.14.5
collected 7 items

test_library_fines.py FFF..F.                                                                          [100%]

================================================== FAILURES ===================================================
_____________________________________ test_parse_member_id_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 8, in test_parse_member_id_no_crash
  |     def test_parse_member_id_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 3 distinct failures. (3 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 10, in test_parse_member_id_no_crash
    |     parse_member_id(s)
    |   File "D:\SE-Lab\Lab7\library_fines.py", line 3, in parse_member_id
    |     return member_type.upper(), int(number)
    |                                 ^^^^^^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 12, in test_parse_member_id_no_crash
    |     assert False, f"parse_member_id crashed with: {e!r}"
    | AssertionError: parse_member_id crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_parse_member_id_no_crash(
    |     s='-',
    | )
```

```
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 10, in test_parse_member_id_no_crash
    |     parse_member_id(s)
    |   File "D:\SE-Lab\Lab7\library_fines.py", line 2, in parse_member_id
    |     member_type, number = member_id.split('-')
    |     ^^^^^^^^^^^^^^^^^^^
    | ValueError: not enough values to unpack (expected 2, got 1)
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 12, in test_parse_member_id_no_crash
    |     assert False, f"parse_member_id crashed with: {e!r}"
    | AssertionError: parse_member_id crashed with: ValueError('not enough values to unpack (expected 2, got 1)')
    | assert False
    | Failing test case: test_parse_member_id_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 3 ----------------
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 10, in test_parse_member_id_no_crash
    |     parse_member_id(s)
    |   File "D:\SE-Lab\Lab7\library_fines.py", line 2, in parse_member_id
    |     member_type, number = member_id.split('-')
    |                           ^^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 12, in test_parse_member_id_no_crash
    |     assert False, f"parse_member_id crashed with: {e!r}"
    | AssertionError: parse_member_id crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_parse_member_id_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________ test_total_fine_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 16, in test_total_fine_no_crash
  |     def test_total_fine_no_crash(s):
  |                    ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 18, in test_total_fine_no_crash
    |     total_fine(s)
    |   File "D:\SE-Lab\Lab7\library_fines.py", line 6, in total_fine
    |     days = [int(d) for d in days_csv.split(',')]
    |            ^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^^
    |   File "D:\SE-Lab\Lab7\library_fines.py", line 6, in <listcomp>
    |     days = [int(d) for d in days_csv.split(',')]
    |             ^^^^^^
    | ValueError: invalid literal for int() with base 10: ''
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 20, in test_total_fine_no_crash
    |     assert False, f"total_fine crashed with: {e!r}"
    | AssertionError: total_fine crashed with: ValueError("invalid literal for int() with base 10: ''")
    | assert False
    | Failing test case: test_total_fine_no_crash(
    |     s='',  # or any other generated value
    | )
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 18, in test_total_fine_no_crash
    |     total_fine(s)
    |   File "D:\SE-Lab\Lab7\library_fines.py", line 6, in total_fine
    |     days = [int(d) for d in days_csv.split(',')]
    |                            ^^^^^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 20, in test_total_fine_no_crash
    |     assert False, f"total_fine crashed with: {e!r}"
```

```
    | AssertionError: total_fine crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_total_fine_no_crash(
    |     s=None,  # or any other generated value
    | )
    +------------------------------------
_____________________________________ test_shelf_code_no_crash _____________________________________
  + Exception Group Traceback (most recent call last):
  |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 24, in test_shelf_code_no_crash
  |     def test_shelf_code_no_crash(s):
  |                   ^^^
  | ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-exceptions)
  +-+---------------- 1 ----------------
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 26, in test_shelf_code_no_crash
    |     shelf_code(s)
    |   File "D:\SE-Lab\Lab7\library_fines.py", line 15, in shelf_code
    |     return (words[0][0] + words[1][0]).upper()
    |             ~~~~~^^^
    | IndexError: list index out of range
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 28, in test_shelf_code_no_crash
    |     assert False, f"shelf_code crashed with: {e!r}"
    | AssertionError: shelf_code crashed with: IndexError('list index out of range')
    | assert False
    | Failing test case: test_shelf_code_no_crash(
    |     s='',
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SE-Lab\Lab7\test_library_fines.py:27
    +---------------- 2 ----------------
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 26, in test_shelf_code_no_crash
    |     shelf_code(s)
    |   File "D:\SE-Lab\Lab7\library_fines.py", line 14, in shelf_code
    |     words = title.split()
    |             ^^^^^^^^^^^
    | AttributeError: 'NoneType' object has no attribute 'split'
    |
    | During handling of the above exception, another exception occurred:
    |
    | Traceback (most recent call last):
    |   File "D:\SE-Lab\Lab7\test_library_fines.py", line 28, in test_shelf_code_no_crash
    |     assert False, f"shelf_code crashed with: {e!r}"
    | AssertionError: shelf_code crashed with: AttributeError("'NoneType' object has no attribute 'split'")
    | assert False
    | Failing test case: test_shelf_code_no_crash(
    |     s=None,
    | )
    | Explanation:
    |     These lines were always and only run by failing test cases:
    |         D:\SE-Lab\Lab7\test_library_fines.py:27
    +------------------------------------
_____________________________ test_total_fine_rejects_negative_days _____________________________

    @given(st.lists(st.integers(min_value=0, max_value=365)),
>          st.integers(min_value=-365, max_value=-1))
                ^^^

test_library_fines.py:48:
- - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - - -

days = [], negative_day = -1

    @given(st.lists(st.integers(min_value=0, max_value=365)),
           st.integers(min_value=-365, max_value=-1))
    def test_total_fine_rejects_negative_days(days, negative_day):
        days_csv = ",".join(str(d) for d in days + [negative_day])
>       assert total_fine(days_csv) is None
E       AssertionError: assert 0 is None
E        +  where 0 = total_fine('-1')
E       Failing test case: test_total_fine_rejects_negative_days(
E           # The test always failed when commented parts were varied together.
E           days=[],  # or any other generated value
E           negative_day=-1,  # or any other generated value
E       )
```

```
test_library_fines.py:51: AssertionError
=========================================== short test summary info ===========================================
FAILED test_library_fines.py::test_parse_member_id_no_crash - ExceptionGroup: Hypothesis found 3 distinct failures. (3
sub-exceptions)
FAILED test_library_fines.py::test_total_fine_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-
exceptions)
FAILED test_library_fines.py::test_shelf_code_no_crash - ExceptionGroup: Hypothesis found 2 distinct failures. (2 sub-
exceptions)
FAILED test_library_fines.py::test_total_fine_rejects_negative_days - AssertionError: assert 0 is None
========================================= 4 failed, 3 passed in 10.29s ========================================
PS D:\SE-Lab\Lab7>
```

Step 5: Run the fixed application

Same inputs as Step 2. Valid input gives the same results; invalid input now prints a clear message instead of crashing.

```
PS D:\SE-Lab\Lab7> Get-Content inputs\valid.txt; cmd /c "python library_fines.py < inputs\valid.txt"; Get-Content
inputs\invalid.txt; cmd /c "python library_fines.py < inputs\invalid.txt"
STU-1023
10,20,30
Data Structures
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): Member Type and Number: ('STU', 1023)

==== Running total_fine ====
Enter days kept for each book as CSV (e.g. 10,20,30): Total Fine: 44

==== Running shelf_code ====
Enter a book title with at least two words: Shelf Code: DS
STU1023
10,,30
Python
==== Running parse_member_id ====
Enter a member ID (e.g. STU-1023): Invalid member ID. Use TYPE-NUMBER, e.g. STU-1023

==== Running total_fine ====
Enter days kept for each book as CSV (e.g. 10,20,30): Invalid input. Enter whole numbers of days separated by commas

==== Running shelf_code ====
Enter a book title with at least two words: Invalid title. Enter at least two words
```

Step 6: Re-run the Hypothesis tests on the fixed code

pytest test_library_fines.py - all 7 tests pass.

```
PS D:\SE-Lab\Lab7> pytest test_library_fines.py
=========================================== test session starts ===========================================
platform win32 -- Python 3.11.9, pytest-9.2.0, pluggy-1.7.0
rootdir: D:\SE-Lab\Lab7
plugins: hypothesis-6.170.15, platformdirs-4.14.5
collected 7 items

test_library_fines.py .......                                                                        [100%]

============================================ 7 passed in 1.65s ============================================
PS D:\SE-Lab\Lab7>
```